

STA 131A – Homework 7

Submission due: Tue, June 2 at 11:59 PM PT

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Instructions: Upload a single PDF file to Gradescope via Canvas (“Homework 7” under “Assignments”). Name the file using the prefix of your UC Davis email ID and the homework number (e.g., `dgsong_hw7.pdf`). Include “STA 131A,” your name, and the last four digits of your student ID on the front page. No late submissions will be accepted; any submission received after the deadline will receive 0 points. For full information about submission requirements and the late submission policy, see the syllabus.

Problem 1 (20 points in total). Sum of a random number of random variables

(a) (10 points) Let

$$S = \sum_{i=1}^N X_i,$$

where $N \in \{0, 1, 2, \dots\}$ is independent of the i.i.d. random variables $X_1, X_2, \dots$

(i) Suppose

$$\mathbb{E}[N] = 4, \quad \text{Var}(N) = 3, \quad \mathbb{E}[X_i] = 2, \quad \text{Var}(X_i) = 5.$$

Compute $\mathbb{E}[S]$ and $\text{Var}(S)$. In one or two sentences, explain how both the randomness of the summands X_i and the randomness of the count N contribute to $\text{Var}(S)$.

(ii) Suppose

$$N \sim \text{Poisson}(\lambda), \quad X_i \sim \text{Bernoulli}(p),$$

with the same independence assumptions as above. Use MGFs to show that

$$S \sim \text{Poisson}(\lambda p).$$

(Hint: $M_N(t) = \exp\{\lambda(e^t - 1)\}$ and $M_{X_i}(t) = 1 - p + pe^t$.)

(b) (10 points) [BT08, Chapter 4, Problem 41]

Problem 2 (30 points in total). Markov and Chebyshev inequalities

(a) (9 points) Suppose $X \geq 0$, $\mathbb{E}[X] = 20$, and $\text{Var}(X) = 64$.

(i) Use Markov’s inequality to bound $P(X \geq 50)$.

(ii) Use Chebyshev’s inequality to bound $P(X \geq 50)$.

(iii) Which bound is smaller at $t = 50$? More generally, for $t > 20$, compare the Markov bound with the Chebyshev bound to determine for which values of $t > 20$ the Chebyshev bound is smaller.

(b) (5 points) Let $X_1, \dots, X_n$ be i.i.d. with

$$\mathbb{E}[X_i] = \mu, \quad \text{Var}(X_i) = 9.$$

Let $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$. Find a sufficient condition on the sample size n to guarantee

$$P(|\bar{X}_n - \mu| \geq 0.2) \leq 0.05$$

using Chebyshev's inequality.

(c) (6 points) Let $A_1, \dots, A_n$ be events, and let $\mathbf{1}_{A_i}$ denote the indicator random variable of A_i :

$$\mathbf{1}_{A_i} = \begin{cases} 1, & \text{if } A_i \text{ occurs,} \\ 0, & \text{otherwise.} \end{cases}$$

Define

$$Z = \sum_{i=1}^n \mathbf{1}_{A_i}.$$

Thus Z is the number of events among $A_1, \dots, A_n$ that occur.

(i) Compute $\mathbb{E}[Z]$.

(ii) Use Markov's inequality to show that

$$P\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n P(A_i).$$

(Hint: the event $\bigcup_{i=1}^n A_i$ is the same as the event $\{Z \geq 1\}$. This inequality is widely known as the *union bound*.)

(d) (10 points) [BT08, Chapter 5, Problem 1]

Problem 3 (25 points in total). Weak law of large numbers

(a) (6 points) Let

$$Y_n = \begin{cases} n, & \text{with probability } 1/n, \\ 0, & \text{with probability } 1 - 1/n. \end{cases}$$

(i) Prove that $Y_n \xrightarrow{P} 0$.

(ii) Compute $\mathbb{E}[Y_n]$. Does $\mathbb{E}[Y_n] \rightarrow 0$?

(iii) Briefly explain what this example shows about whether convergence in probability necessarily implies convergence of expectations.

(b) (6 points) Let $X_1, X_2, \dots$ be i.i.d. with

$$\mathbb{E}[X_i] = \mu, \quad \text{Var}(X_i) = \sigma^2 < \infty.$$

Let $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$. Use Chebyshev's inequality to prove that

$$\bar{X}_n \xrightarrow{P} \mu.$$

- (c) **(5 points)** Let $X_1, \dots, X_n$ be i.i.d. Bernoulli(p), and let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Use Chebyshev's inequality to bound $P(|\bar{X}_n - p| \geq \epsilon)$. Then find a sufficient condition on n , possibly depending on p , that guarantees

$$P(|\bar{X}_n - p| \geq 0.05) \leq 0.01.$$

Finally, find a sufficient condition on n that guarantees the same inequality for every $p \in [0, 1]$.

- (d) **(8 points)** [BT08, Chapter 5, Problem 5]

Problem 4 (25 points in total). Central limit theorem

- (a) **(5 points)** Let $X_1, \dots, X_n$ be i.i.d. with

$$\mathbb{E}[X_i] = \mu, \quad \text{Var}(X_i) = \sigma^2 < \infty \quad (\sigma > 0).$$

Let

$$S_n = X_1 + \dots + X_n, \quad \bar{X}_n = \frac{S_n}{n}.$$

Write the CLT approximations for

$$P(S_n \leq c) \quad \text{and} \quad P(\bar{X}_n \leq a),$$

expressing your answers in terms of the standard normal CDF Φ .

- (b) **(6 points)** Suppose $X_1, \dots, X_{64}$ are i.i.d. Exponential(2), where 2 is the rate parameter. Let

$$\bar{X}_{64} = \frac{1}{64} \sum_{i=1}^{64} X_i.$$

Use the CLT to approximate

$$P(0.45 \leq \bar{X}_{64} \leq 0.55).$$

(Recall: if $X \sim \text{Exponential}(\lambda)$, then $\mathbb{E}[X] = 1/\lambda$ and $\text{Var}(X) = 1/\lambda^2$.)

- (c) **(6 points)** Let

$$S \sim \text{Binomial}(100, 0.4).$$

Use the normal approximation to approximate

$$P(35 \leq S \leq 45).$$

Express your answer using Φ , and then give a numerical approximation.

- (d) **(8 points)** [BT08, Chapter 5, Problem 8]

Problem 5* (10 bonus points). Bonus problems

- (a) **(5 bonus points)** [BT08, Chapter 4, Problem 44]

- (b) **(5 bonus points)** [BT08, Chapter 5, Problem 10]

References

- [BT08] Dimitri Bertsekas and John N Tsitsiklis. *Introduction to probability*, volume 1. Athena Scientific, 2nd edition, 2008.